

VENKATA RAO GADDAM

PRINCIPAL / CHIEF EMBEDDED SYSTEMS ARCHITECT

BARE-METAL • ANDROID BSP • LINUX BSP • RTOS • QNX • DIGITAL COCKPITS • ADAS • AI/ML

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BANGALORE, INDIA

EXECUTIVE PROFILE

Principal / Chief Embedded Systems Architect and technology leader with **22+ years of experience** in embedded software architecture, automotive software, real-time systems, Android/Linux/QNX platforms, BSP development, device drivers, system integration, and product engineering across automotive and other safety-critical domains.

Extensive experience architecting and delivering **automotive Digital Cockpit, IVI, Instrument Cluster/METER, HUD, ADAS and AI/ML systems** across Qualcomm, Samsung, NXP, MediaTek and Telechips platforms. Strong expertise in **Android BSP, Android Internals, Android HAL/NDK, Android OS migrations, Embedded Linux, Linux Kernel, QNX, hypervisors, board bring-up, platform customization and device-driver development.**

Hands-on experience with Qualcomm automotive platforms including **SA8155P, SA8255P, SA8295P, SA8775P, SA7255P**, Telechips **D3/D5**, and Samsung **Exynos Auto V920** platforms. Delivered Android and Linux platform migrations including **Android 12/13/14/15 and Linux Kernel 5.4 to 6.1.**

Experienced in **AI/ML model development and deployment for ADAS**, including YOLO26n, YOLO11n, MediaPipe Face Mesh, Dlib ResNet34/50, QNN SDK and HTP-based inference. Proven technical leadership across architecture, design reviews, RFQ/SOW support, estimation, customer engagement, team building, mentoring, delivery management and complex product development.

CORE COMPETENCIES

- Embedded Systems Architecture & Product Engineering | Automotive Software Architecture | Digital Cockpit Architecture | Android Automotive / Android BSP | Android Internals | Android HAL / Android NDK | Android OS Migration & Platform Upgrades | Embedded Linux / Linux Kernel / Linux BSP | QNX BSP & Application Development | Non-hypervisor & Hypervisor-based Platforms | Qualcomm Automotive Platforms | Board Bring-up & Platform Customization, Device Drivers & Hardware/Software Integration | IVI / Instrument Cluster / METER / HUD | ADAS / DMS / OMS / SVM | AI/ML Model Development & Deployment | QNN / CPU / GPU / DSP / HTP / NPU Inference | x86 / ARM / RISC-V Embedded Platforms | BootROM / Bootloader / U-Boot / OpenSBI | Functional Safety & Cybersecurity | System Architecture & Design Reviews
- Design, development, and delivery of complex embedded products across automotive, industrial automation, medical, railway, process control, security, energy & utilities, telecom, pharma, and storage domains
- Led a 100+ member global embedded engineering organization to deliver mission-critical firmware and hardware platforms, optimizing resource allocation across multiple concurrent product lines
- Specialized in architecting and writing firmware for various 8/16/32/64-bit microcontrollers and microprocessors architectures
- Built and scaled embedded/BSP engineering organizations, Managed engineering budgets, resource allocation and delivery commitments
- Technical Strategy / Architecture Reviews / RFQ & SOW / Estimation / Team Building / Mentoring / RCA & Issue Resolution

TECHNICAL SKILLS

Programming: C, C++, Python, Assembly, Shell script, LabVIEW

Operating systems & RTOS: Embedded Android, Android Automotive (AAOS), Embedded Linux, QNX, Zephyr, FreeRTOS, ThreadX, VxWorks, uCOS/OS, RTX-51, WinCE, mbed

Automotive & Embedded Platforms: Qualcomm SA8155P, SA8295P, SA8255P, SA8775P SA7255P, Samsung Exynos Auto V920, Telechips D3/D5, NXP i.MX95, MediaTek, ARM Cortex, RISC-V, x86, Microchip, ATMEL AVR, TI MSP430

Android / Linux / BSP: Android BSP, Android Internals, Android HAL, Android NDK, Android Porting, Android OS Upgrades, Linux BSP, Linux Kernel, Linux Porting, Device Drivers, Root Filesystems, GKI, Yocto, OpenEmbedded, Buildroot, BusyBox

Hypervisors: Gunyah, CoQOS, QNX Hypervisor, Xen

AI/ML: YOLO11n, YOLO26n, MediaPipe Face Mesh, Dlib ResNet50/34, Model training/validation/testing, QNN SDK, CPU/GPU/DSP/HTP inference

Deep Learning & Computer Vision: Scikit-learn, TensorFlow/Keras, PyTorch, YOLO, OpenCV, Torchvision, CNN, ANN, Transformers, NLP/GenAI

AI/ML Tools: Roboflow, LabelMe, LabelImg, TensorBoard, Netron, Jupyter Notebook, Google Colab, Saturn Cloud, Databricks

Build & DevOps: Git, Gerrit, GitHub, GitLab, Jenkins, Docker, CMake, Yocto, OpenEmbedded, Buildroot, BusyBox

Debug & Quality: GDB, KGDB, KDB, Lauterbach, OpenOCD, ADB, QFIL, QPST, QXDM, Coverity, Klocwork, QAC, LDRA, VectorCAST, CANoe

Interfaces & Protocols: UART, Modbus, I2C, SPI, I3C, I2S, TDM, PCM, QSPI, CSI, DSI, Ethernet, TCP/IP, TLS, USB, CAN, CAN-FD, LIN, RS232/485/422, BLE, Wi-Fi, SCMI, Zigbee, LoRa, NB-IoT, NFC, Wirepas, SOME/IP

Standards: ISO 26262, ISO/SAE 21434, UNECE WP.29, IEC 62304, IEC 60601-1, ISO 13485, ISO 14971, IEC 61508, MISRA C/C++

Other Tools: VS Code, Python CLI, QNX Momentics, eclipse, Logic Analyzers, Oscilloscopes, Spectrum Analyzer

PROFESSIONAL EXPERIENCE

Senior Solution Architect – Automotive Embedded Software | KPIT Technologies Ltd., Bangalore | Sep 2022 – Present

- Provide **technical architecture, system design reviews, effort estimation, project planning, issue resolution, and engineering leadership** for automotive digital cockpit programs.
- Lead **BSP development and customization, Android/Linux porting, platform drivers, Android OS upgrades, Linux Kernel upgrades, security updates and feature development** for Qualcomm, Samsung, NXP, MediaTek and Telechips automotive platforms.
- Lead architecture and development for **Digital Cockpit systems including IVI, METER/Instrument Cluster, and HUD**, while building and mentoring Android, Linux, and QNX engineering teams.
- Drive **AI/ML product development** for Digital Cockpits, ADAS and AD use cases.
- Architected and developed an **Advanced Driver Monitoring System (DMS)** on Qualcomm SA8775P covering model selection, dataset preparation, YOLO11n custom model development, MediaPipe Face Mesh, Dlib ResNet34, model conversion and QNN/HTP deployment.
- Delivered **Android 12/13/14/15 and Linux kernel 5.4 to 6.1 migrations** across SA8155P, SA8295P, SA8255P, SA7255P, SA8775P, i.MX 95, Exynos Auto V920, Telechips D3 and D5 platforms.
- Developed DMS applications using **Linux/Python, QNX/C++, and Android/NDK**; performed SA8775P Ride SX 4.0 HQX platform bring-up and camera integration.
- Established KPIT DevOps setup for OEM services, contributed to building BSP engineering teams, and supported development of the Qualcomm Authorized Design Center team.

Senior Staff Engineer – Embedded Software | Ceremorphic Technologies Pvt. Ltd., Hyderabad | Mar 2021 – Mar 2022

- Led embedded software development for **ARM Cortex-M55 and RISC-V SoCs** covering BootROM, bootloader, RTOS/Linux porting, platform drivers, FPGA pre-silicon testing, and engineering management.
- Built ARM/RISC-V toolchains; integrated Eclipse/QueStudio IDE support; completed BootROM evaluation on FPGA/ASIC platforms.
- Ported Linux to RISC-V using **OpenSBI and U-Boot** and customized OpenSBI, U-Boot, Linux Kernel, and root filesystems.
- Completed 32/64-bit RISC-V based SoC design activities supporting **functional safety and ISO 26262 ASIL-D certification**.

Senior Technical Manager – Embedded Software | Votary Softech Solutions Pvt. Ltd., Hyderabad | Aug 2020 – Feb 2021

- Managed customer and vendor relationships, engineering resources, delivery activities and BSP customization.
- Led **Android/Linux porting on Qualcomm mobile chipsets**, OSS customization, root filesystem development, vulnerability management, and device-driver development.
- Developed reusable technology frameworks, coding/design guidelines, and a **C runtime library for performance improvement**.

Senior Technical Manager – Embedded Systems | Secure Meter Limited, Udaipur | Nov 2017 – Jun 2020

- Led end-to-end design and firmware development for **7+ advanced industrial instrumentation and smart metering platforms** using ARM Cortex processors including STM32, NXP, SiLabs and i.MX6.
- Integrated **Modbus RTU, BLE, Wirepas Smart Mesh and IEC 60870-5-104** for contactless utility billing, substation automation and high-accuracy DUT current profiling.
- Implemented precision analog front ends (AFEs), external 24-bit ADCs/DACs, and closed-loop control logic for calibration-free sensing and thermodynamic pump-efficiency monitoring.
- Completed **PDPS firmware development in 7 days** with process documentation and received Best Mentor Award.

Engineering Technical Manager – Embedded Systems | eInfoChips Limited, Ahmedabad | Jun 2016 – May 2017

- Led **SOW preparation, estimation, planning, software architecture, device-driver development, BSP customization, Android/Linux porting, technical reviews, and customer management**.
- Engineered dual-camera event recorders using vehicle OBD dashboard telemetry to trigger automated video recording events.
- Developed an ARM-based Portable Navigation Device and architected an IoT sensor node for smart streetlights using DALI communication.
- Built and customized Embedded OS distributions including **Yocto, OpenEmbedded, Ubuntu Core, and Android 5.1.1/6.0.1** on Qualcomm APQ8016, APQ8096, APQ8064.
- Designed a production-ready **MFG Test Suite in LabVIEW** for automated high-volume manufacturing testing and precision meter calibration.

Engineering Technical Manager – Embedded Software | HCL Technologies Limited, Chennai | Jul 2014 – Apr 2016

- Led proposals, estimation, software architecture, development, technical reviews, RCA, project/quality/customer management, and mentoring.
- Engineered automated testing and disinfection firmware for the Evotech endoscope system using Linux and real-time digital/analog I/O.
- Architected a **Linux-based ARM Cortex-A9** communication gateway (ARP) connecting Alaris Infusion Pumps with ACE server infrastructure.
- Designed an **ARM Cortex-M4** 2D barcode scanner module integrating image sensors with RS-485/UART interfaces.

Technical Leader – Embedded Systems | General Electric, HTC, Hyderabad | Apr 2010 – Jul 2014

- Owned requirements, software architecture, design, coding, reviews, system integration, performance support, stage gates, estimation, vendor coordination, and technical mentoring.
- Designed and developed high accuracy data acquisition firmware using C/C++ across **ARM-Cortex-A8, MSP430, PIC18 and PIC24H** platforms.
- Developed systems using WinCE, OS9000, ThreadX RTOS for Thermal Validation and Protimeter Products.
- Received 2 management awards and 6 Above & Beyond awards for technical excellence.

Project Lead – Embedded Systems | Sugaan Automatics Pvt. Ltd., Hyderabad | Apr 2009 – Feb 2010

- Led technical/project management and hardware/firmware development for energy meters, AMR, shelter automation, and IDD products.
- Designed and developed firmware using **ARM7 LPC2214/LPC2103** and Wavecom Modem.

Senior Engineer – Embedded Software | Patni Computer Systems Ltd., Mumbai | Nov 2007 – Mar 2009

- Designed and developed embedded software for **railway braking systems & System Redundancy Modules** using C/C++ across ARM9 STR912FW44, PPC440, MCF5272 Coldfire, MC68HC912DG128A and Intel i960Hx platforms.
- Worked with **Linux and VxWorks** embedded operating systems.

Senior Engineer – Embedded Systems | HBL Nife Power Systems Ltd., Hyderabad | Aug 2004 – Oct 2007

- Led software architecture, requirements analysis, design reviews, traceability, system integration, and end-to-end product lifecycle execution for **railway signaling systems**.
- Designed firmware for electronic interlocking, datalogger, panel processor and digital axle counter systems using Assembly/C/C++ and RTX-51, pSOS and uCOS/OS.
- Received 4 management awards for Technical Excellence and Expertise.

R&D Engineer / Senior R&D Engineer | Spectrochem Instruments Pvt. Ltd., Hyderabad | Apr 2000/2003 – Aug 2001/2004

- Designed, developed, and delivered production-ready firmware and hardware layouts for SCADA instrumentation, automatic temperature controllers, smart pH controller, bio-fermenter, and industrial security systems.
- Developed embedded solutions across **8/16/32-bit microprocessors & microcontrollers** including Intel 8085, 8086, 80386, 8051, Atmel, Atmel AVR, Microchip.

EDUCATION

- M.Tech., Automotive Software Engineering** – Coventry University, United Kingdom – 75%
- M.Phil., Electronics and Instrumentation** – Nagarjuna University, Guntur, Andhra Pradesh – 72%
- M.Sc., Electronics** – Nagarjuna University, Guntur, Andhra Pradesh – 67%
- Post Graduate Diploma in Electronic Instrumentation** – Nagarjuna University, Guntur, Andhra Pradesh – 65%
- B.Sc., Electronics** – Nagarjuna University, Guntur, Andhra Pradesh – 69%

CERTIFICATIONS & PROFESSIONAL TRAINING

Certified Functional Safety Engineer – TÜV SÜD | ISO 26262 | ISO/SAE 21434 | IEC 62304 | ISO 13485 | ISO 14971 | IEC 60601-1 | IEC 60870-5-104 and DLMS | Certified GB-DFSS Lean Six Sigma | Certified LabVIEW Professional | UL and EMI/EMC techniques | Medical Device FDA / Regulatory Training

LEADERSHIP & PROFESSIONAL STRENGTHS

Technical Leadership | Product Architecture | Automotive Software Architecture | Customer Management | Team Building | Mentoring | Product Design & Development | System Integration | Fail-Safe System Design | Delivery Management | Technical Reviews | Root Cause Analysis | Process Improvement | Functional Safety | Technical Strategy

PUBLICATIONS & THOUGHT LEADERSHIP

- The Brain on Wheels: AI-Powered Digital Cockpits and the Future of Software-Defined Vehicles** – *International Journal of Innovative Research in Technology (IJIRT)*, Vol. 13, Issue 3, August 2026; Paper ID: IJIRT207394. Focus: AI-Powered Digital Cockpits, SDV architecture, Edge AI, automotive AI workloads, centralized vehicle computing and next-generation automotive intelligence.
- Automotive Phone Projection** – *International Journal of Naval Research and Development (IJNRD)*, Vol. 8, Issue 7, May 2023; Paper ID: IJNRD2307071. Focus: Android Auto, Apple CarPlay, IVI architecture, smartphone integration and automotive connectivity.
- Seamless, Secure Wi-Fi Connectivity for In-Vehicle and V2X Communication** – *International Journal of Naval Research and Development (IJNRD)*, Vol. 8, Issue 7, July 2023; Paper ID: IJNRD2307072. Focus: OpenRoaming, Passpoint/Hotspot 2.0, secure authentication, seamless Wi-Fi connectivity, IVI, and V2X communication.